

Batch App Presentation

Enterprise Service Bus (ESB) Hub

Designed and implemented by Brahamia Oualid

Professional Bulk Transaction Management



Technical Architecture Details

The system follows a modern service-oriented architecture (SOA).



Frontend Implementation

- **React 18 & Vite:** Fast HMR.
- **PrimeReact & Bootstrap:** UI.
- **Context API:** Global State.
- **React-Router:** Secure Routing.



Backend Infrastructure

- **Node.js & Express:** API layer.
- **Oracle Database:** Persistent Audit.
- **Node-Cron:** Background Throttling.
- **Axios:** SOAP/REST Client.



Active Module: Create Batch

Automated subscriber onboarding with high-volume handling capacity.

- **File Engine:** Supports .xlsx and .csv bulk file injection.
- **Input logic:** Automatic MSISDN range validation and normalization.
- **ESB Trigger:** Asynchronous requests to ESB WebServices.
- **Audit:** Every transaction is logged with its unique Trace-ID.



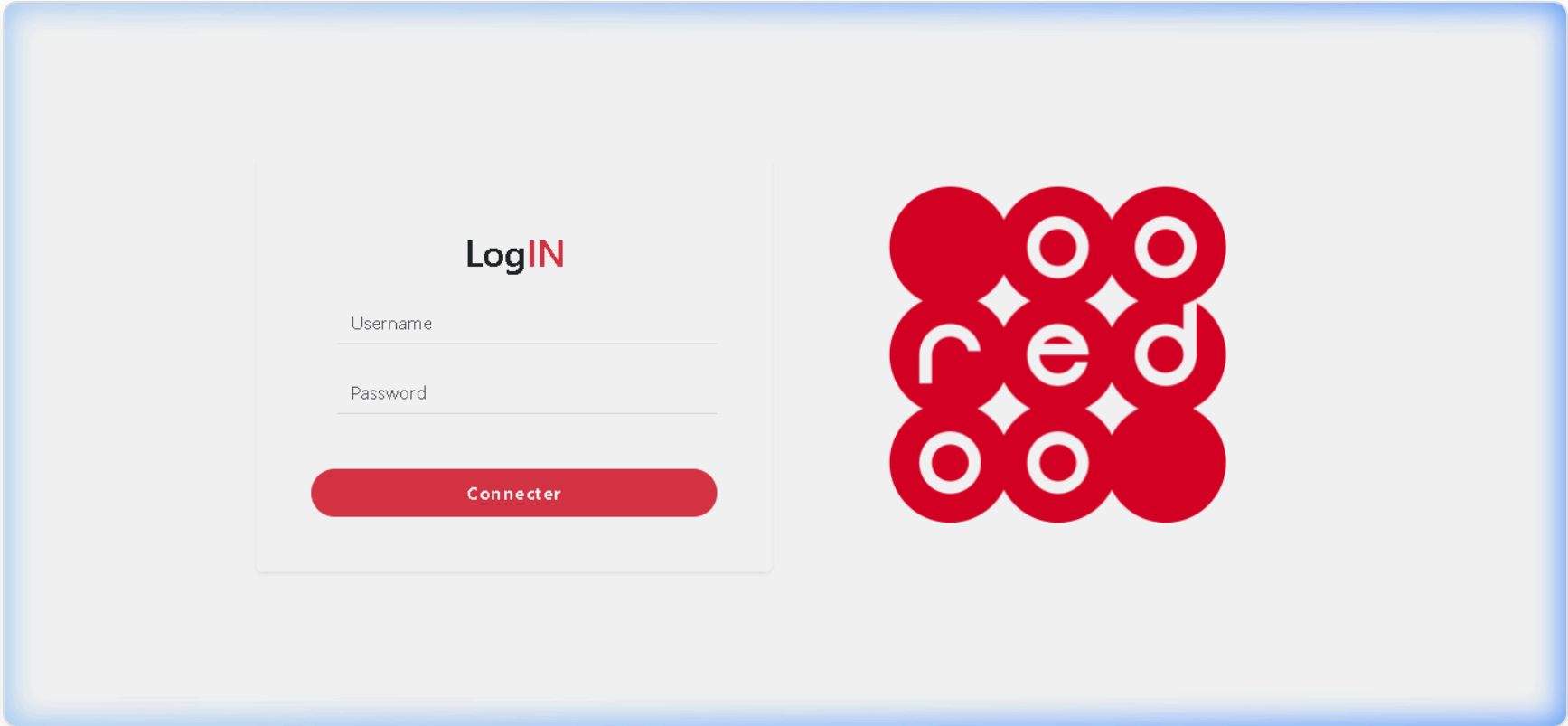
Active Module: Set Contract Status

Bulk management of subscriber lifecycles (Active, Suspend, etc.)

- **State Sync:** Real-time synchronization with HLR/HSS.
- **Flow Control:** **Pause/Resume/Cancel** functionality for job safety.
- **Throttling:** Background workers execute tasks without blocking UI.
- **Persistency:** Batch data stored securely in Oracle for auditing.

1. System Access & Security Gateway

Features JWT-based authentication for verified operator access.



LogIN

Username

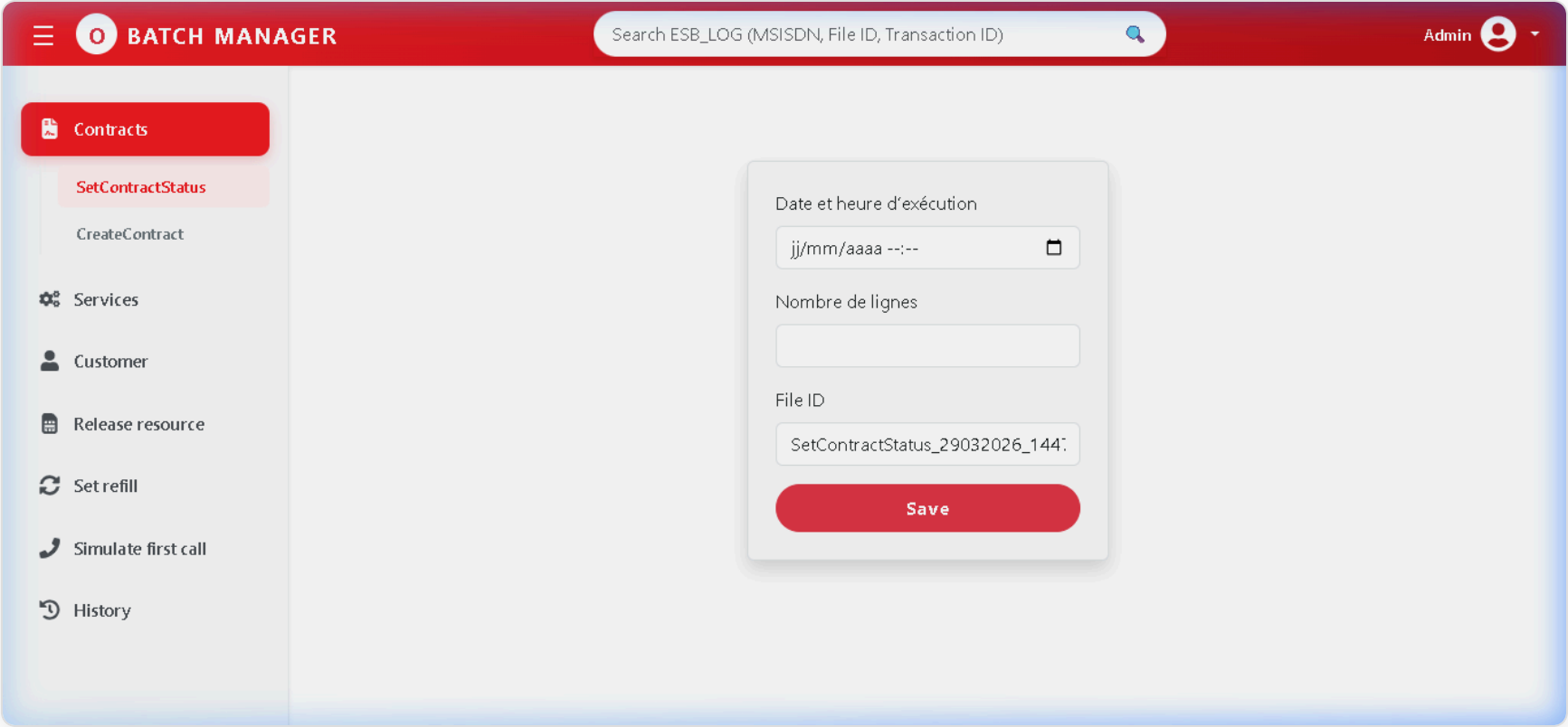
Password

Connector

red

2. Automated Lot Import console

Dynamic console for file upload and real-time operational status.



3. Full Transactional Audit logs

Granular monitoring of each MSISDN with detailed error code descriptions.

BATCH MANAGER

Search ESB_LOG (MSISDN, File ID, Transaction ID)

Admin

Contracts

Services

Customer

Release resource

Set refill

Simulate first call

History

Batch History

← Back

Remove

Batch Details: SetContractAndServices_29032026_142535

SetContractAndServices_29032026_142535

PROCESSED

Operation: CREATE_CONTRACT

Records: 49

Scheduled: Invalid Date

Refresh Data

Export Results

Individual Transaction Results

Showing 49 of 49 total records

MSISDN	Transaction ID	Status	Message	Step	Date
213550013616		FAILED	connect ECONNREFUSED 127.0.0.1:170	-	-
213550013617		FAILED	connect ECONNREFUSED 127.0.0.1:170	-	-
213550013618		FAILED	connect ECONNREFUSED 127.0.0.1:170	-	-

4. Global Search & Audit History

Full historical traceability of thousands of transactions via Oracle DB logs.

BATCH MANAGER

Search ESB_LOG (MSISDN, File ID, Transaction ID)

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Batch Execution History

FILE ID	OPERATION	RECORDS	PROGRESS	UPLOAD DATE	STATUS	ACT
SetContractAndServices_29032026_142535	CREATE_CONTRACT	49	49/49 100%	29/03/2026 14:25:52	PROCESSED	
SetContractAndServices_29032026_141335	CREATE_CONTRACT	49	49/49 100%	29/03/2026 14:13:47	PROCESSED	
SetContractAndServices_29032026_141102	CREATE_CONTRACT	49	49/49 100%	29/03/2026 14:11:20	PROCESSED	
SetContractAndServices_29032026_140127	CREATE_CONTRACT	49	49/49 100%	29/03/2026 14:02:29	PROCESSED	
SetContractAndServices_29032026_135400	CREATE_CONTRACT	49	13/49 27%	29/03/2026 13:54:16	PAUSED	

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BATCH MANAGER

213550013616

✕ 🔍

Admin 

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Batch History



Global Transaction Search

Search directly within the ESB_LOG database by MSISDN, File ID, or Transaction ID.

 213550013616



✨ System Key Highlights

- **Efficiency:** Reduces manual labor by an estimated 95% for bulk tasks.
- **Security:** JWT tokens ensure all batches are attributed to a user.
- **Reliability:** Persistence layer ensures zero data loss on ESB errors.
- **Diagnostics:** Map specific ESB errors (e.g., 500, 404) to user-friendly tips.

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